TEXORA · SESSIONAL NOTES · STUDY SHEET

Wet Wash Processes of Garments

Carrying out and recording the water-based garment wash family — normal/rinse wash, enzyme wash, stone wash, bleach wash and acid wash on denim — with recipes calculated from liquor ratio and load, and the wash effects judged against a buyer wash standard.

AM-4102 · ADVANCED APPAREL MANUFACTURING-I SESSIONAL · STUDY 1 · LEVEL 4 TERM 1

1. Aim and place of the study

Aim: to prepare, run and evaluate the principal wet (aqueous) wash processes applied to finished garments — chiefly denim jeans — calculating each recipe from load and liquor ratio, controlling time, temperature and pH, and assessing the result for shade, hand feel, strength loss and back-staining against an approved wash standard.

Wet washing is done **after sewing**, on the made garment, so it sits between production and finishing. Its purposes are: to remove size, dust and sewing dirt (rinse/normal wash); to soften the hand (enzyme, softener); to create fashion effects — fading, abrasion patterns, vintage looks (stone, bleach, acid wash); and to stabilise dimensions before measurement and packing. Every wet process trades effect against damage: the same chemistry that fades indigo also attacks cellulose, so this study measures both sides of the trade.

2. Materials, machines and specimens

Machines: front-loading garment washing machine (sample machine, 10 kg) with temperature and dosing control; hydro-extractor; tumble dryer; pH meter; balance (1 g); grey scales for colour change and staining (ISO 105-A02/A03); tensile tester access for seam/fabric strength comparison; spray cabinet and PPE for the acid-wash trial (instructor-supervised).

Specimens: five identical pairs of rigid (unwashed) indigo denim jeans from one production lot, plus one retained unwashed pair as the reference standard. One pair per process: (A) rinse wash, (B) enzyme wash, (C) stone–enzyme wash, (D) bleach wash, (E) acid wash. Chemicals: desizing enzyme (alpha-amylase), cellulase enzyme (neutral), pumice stones, sodium hypochlorite bleach, sodium metabisulphite (anti-chlor), hydrogen peroxide, detergent, acetic acid, silicone/cationic softener, potassium permanganate (for E, applied as spray in the full-scale process; in this study a small pre-soaked pumice trial is used instead — see precautions).

3. The process family and its mechanisms

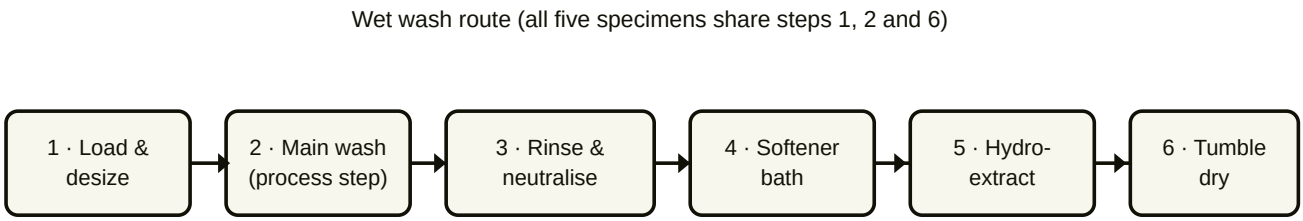


Figure 1 — The common wet-wash route. Only the main-wash chemistry changes between the five specimens.

PROCESS	ACTIVE AGENT AND MECHANISM	EFFECT PRODUCED	MAIN RISKS
Rinse / normal wash	Detergent and water only; removes size (if desize included), dirt and loose dye.	Clean, slightly softer garment; shade nearly unchanged; shrinkage stabilised.	Creasing if overloaded; uneven softening without desize.
Enzyme wash	Cellulase hydrolyses surface cellulose fibrils; loosened indigo-bearing fibres are removed by machine action.	Surface fading (bio-polishing effect), softer hand, less pilling.	Strength loss if overdosed/overlong; back-staining of pocketing and labels.
Stone wash	Pumice stones abrade the garment surface mechanically; usually combined with cellulase (stone–enzyme).	Strong, uneven vintage fading, abraded edges and seams.	Stone dust in pockets, machine wear, heavy strength loss, high water/solid waste.
Bleach wash	Sodium hypochlorite oxidises indigo to colourless products; dose controls the shade drop.	Controlled overall lightening, from mid-stone to pale sky shades.	Chlorine damage to cellulose and elastane; yellowing if not neutralised (anti-chlor step).
Acid wash	Pumice soaked in potassium permanganate (or bleach) discharges indigo only where the stones touch.	Sharp white–blue mottled ("marble/snow") contrast.	Localised severe damage, permanganate stains, strict chemical handling rules.

4. Recipe calculation — worked example

WORKED RECIPE — STONE–ENZYME WASH, SPECIMEN C

Load = **60 kg** of jeans (scaled batch figure for calculation practice; the sample machine runs 6 kg at the same ratios). Liquor ratio **1 : 10** → bath volume = 60 × 10 = **600 L**. Doses: detergent **2 g/L** → 600 × 2 = 1,200 g = **1.2 kg**. Neutral cellulase **0.8% owg** (on weight of goods) → 60 × 0.008 = **0.48 kg**. Pumice **0.5 kg per kg of garment** → 60 × 0.5 = **30 kg**. Conditions: pH 6–7 (neutral cellulase), 50 °C, 45 min. Note the two dosing languages in one recipe — g/L scales with the *bath*, % owg scales with the *goods*; confusing them is the commonest recipe error in the wash plant.

5. Procedure

1. **Desize (all specimens):** alpha-amylase 1.5 g/L, detergent 1 g/L, LR 1:10, 60 °C, 15 min; hot rinse, cold rinse. Record the bath data on the wash card.
2. **Specimen A — rinse wash:** detergent 1 g/L, 40 °C, 20 min. Rinse twice.
3. **Specimen B — enzyme wash:** neutral cellulase 1% owg, pH 6–7, 50 °C, 45 min; enzyme kill: raise to 75 °C for 10 min or shift pH above 9.
4. **Specimen C — stone–enzyme:** the worked recipe of §4; remove stones, rinse until the drain water runs free of grit.
5. **Specimen D — bleach wash:** after enzyme pre-wash (20 min), drain; fresh bath, sodium hypochlorite dosed to the target shade (record g/L of available chlorine), 45 °C, 20 min, checking shade every 5 min against the standard under the light box; anti-chlor with sodium metabisulphite 2 g/L, 10 min — residual chlorine must test negative.
6. **Specimen E — acid wash trial:** pumice pre-soaked in potassium permanganate solution by the instructor; tumble the dry garment with the treated stones without liquor for 20 min; neutralise (metabisulphite/oxalic bath) until the brown MnO_2 staining clears, then wash off.
7. **Common finish:** softener bath (cationic softener 1% owg, pH 5.5 with acetic acid, 35 °C, 10 min), hydro-extract 3 min, tumble dry at 70 °C to bone dry, condition 4 h before evaluation.
8. **Evaluation:** grade colour change and back-staining with the grey scales; compare hand and appearance with the reference pair; test grab tensile strength of a warp strip from each specimen against the unwashed control and record % strength loss.

Observation points — what to record

- Full bath data per specimen: load, LR, doses (both languages), temperature, time, pH start/end.

- Grey-scale grades for colour change and for staining of pocketing/labels (back-staining), per specimen.
- Strength loss % per process, and the rank order of the five effects from darkest to lightest.
- Defects: crease marks, stone damage, uneven patches, residual chlorine test result for D.
- Water and chemical totals per kg of garment, from the batch figures.

Precautions

- Hypochlorite and acid must never meet — chlorine gas. Bleach baths and acid (acetic) dosing are kept in separate steps with a rinse between, as in the procedure.
- Permanganate handling, soaking and disposal are instructor-only operations; gloves and eye protection for all chemical handling.
- Kill the enzyme at the end of its step: active cellulase left in a warm bath keeps eating cellulose during delays.
- Never judge shade on a wet or warm garment — fading is assessed dry, conditioned, under standard light.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why must rigid denim be desized before enzyme or bleach washing?
2. A batch of 80 kg is washed at LR 1:8 with detergent at 1.5 g/L and cellulase at 1.2% owg. Calculate the bath volume and both chemical quantities.
3. Distinguish the fading mechanisms of enzyme wash, stone wash and bleach wash.
4. What is back-staining, in which specimens is it worst, and how is it reduced?
5. Why is an anti-chlor step compulsory after hypochlorite bleaching, and how do you verify it worked?

Answer key

1. Warp size (starch) coats the yarn and blocks uniform access of enzyme/bleach to the indigo-bearing surface; without desizing the wash is patchy and the effect cannot be reproduced batch to batch.
2. Bath = 80 × 8 = 640 L. Detergent = 640 × 1.5 = 960 g. Cellulase = 80 × 0.012 = 0.96 kg. (Note the near-equal numbers are coincidence — the bases differ: bath vs goods.)
3. Enzyme: biochemical hydrolysis of surface cellulose, removing dye-bearing fibrils. Stone: purely mechanical abrasion by pumice. Bleach: chemical oxidation destroying the indigo chromophore itself — it lightens even without fibre removal.

4. Indigo removed in the bath redeposits on white/undye d surfaces — pocketing, labels, weft yarns — greying them. Worst in enzyme and stone washes (most loose dye in the bath). Reduced by anti-back-staining agents, shorter baths, prompt rinsing and keeping pocketing quality in mind at product development.
5. Residual active chlorine continues to oxidise cellulose (and elastane) in storage, causing strength loss and yellowing. Metabisulphite reduces it to harmless chloride; verification is a negative residual-chlorine test (e.g. potassium iodide–starch paper shows no blue) before the garment leaves the wash.
